Lesson 5: Induction

Only changing linked flux produces the pulse

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

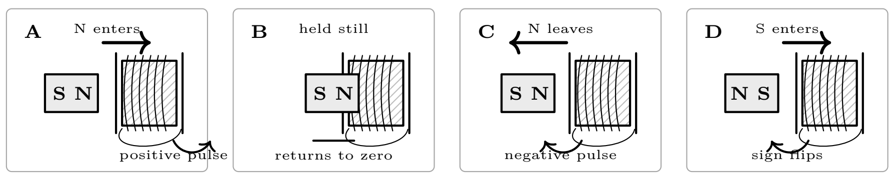
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- Coil of 200+ turns
- Bar magnet
- Sensitive multimeter
- Cardboard guide tube

Predict first

Predict the meter sign for N-pole in, N-pole out, S-pole in, and S-pole out. Which states should return to zero?



Meter-lead convention fixes which deflection is called +.

same path and speed

Investigate

1. With nothing powered, connect the coil to the meter's voltage input on its lowest suitable DC range. Fix the lead colors and coil orientation for every trial. Zero the display. *Does touching neither lead produce zero? Does moving the cable alone create a false pulse?*
2. Mark start, center, and finish positions on the cardboard guide. Hold the N pole still at each mark for three seconds. Record the reading after it settles. *Is location alone sufficient, or must linked flux change?*
3. Push N from start to finish on a slow three-count. Record peak value and sign. Repeat three times without changing grip or path; report the median and range. *Were the three peaks close enough to compare?*
4. Repeat the N-in trial on a fast one-count. Then pull N out at the same two speeds, three trials each. *Which variable changed magnitude? Which change reversed sign? Does the meter return to zero after every stop?*
5. Turn the magnet end-for-end without swapping meter leads. Repeat the matched S-in and S-out trials. Before each movement, state the predicted sign. *Does reversing the pole reproduce the same sign change as reversing motion? What happens if both reverse?*
6. Final verification: select one speed and perform N-in, N-out, S-in, S-out twice in that order. Accept the result only if each repeated pair agrees in sign and the stationary readings return near zero.
7. If readings disagree, stop and check loose terminals, wrong meter jack/range, changed coil orientation, side-to-side magnet motion, inconsistent speed, magnetic objects near the guide, or auto-ranging delay. Correct one cause, then repeat the full verification set.

Explain from the evidence

The meter responds to induced emf, which follows the rate of change of magnetic flux linked through the coil. A magnet can produce strong flux while held inside yet no sustained emf because the linkage is no longer changing. Faster motion produces a larger rate of change; additional linked turns generally increase the peak. Reversing motion reverses the change, and reversing the pole reverses the flux direction. Reversing either one reverses meter sign; reversing both preserves it. The induced current's field opposes the imposed change—Lenz's law—so mechanical work supplies the measured electrical energy. A sign is meaningful only after meter leads and coil orientation are fixed.

Change one thing

Make a four-row table for N-in, N-out, S-in, and S-out with predicted sign, two measured signs, median peak, and stationary final reading. Then drop the magnet through the nonconducting guide from one marked height three times. Sketch the positive and negative pulses in time order, identify entry and exit, and explain any unequal peaks from changing speed rather than claiming a violation of the sign rule.

Evidence record	
Prediction	______________________
One change	______________________
Observation	______________________
Claim + because	______________________

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.